Ecological Informatics 74 (2023) 101991

Contents lists available at ScienceDirect

Ecological Informatics

journal homepage: www.elsevier.com/locate/ecolinf

Predicting Water Quality Index (WQI) by feature selection and machine learning: A case study of An Kim Hai irrigation system

Bui Quoc Lap [a,*], Thi-Thu-Hong Phan [b,*], Huu Du Nguyen [c,d], Le Xuan Quang [e], Phi Thi Hang [e], Nguyen Quang Phi [f], Vinh Truong Hoang [g], Pham Gia Linh [h], Bui Thi Thanh Hang [h]

[a] Faculty of Chemistry & Environment, ROOM team, Thuy loi university, 175 Tay Son, Dong Da, Hanoi, Viet Nam
[b] Department of Artificial Intelligent, FPT University, Danang, Viet Nam
[c] School of Applied Mathematics and Informatics, Hanoi University of Sciences and Technology, Hanoi, Viet Nam
[d] Department of Mathematics, Faculty of Information Technology, Vietnam National University of Agriculture, Hanoi, Viet Nam
[e] Institute for Water and Environment, Vietnam Academy for Water Resources, 171 Tay Son, Dong Da, Hanoi, Viet Nam
[f] Faculty of Water Resources Engineering, Thuyloi University, 175 Tay Son, Dong Da, Hanoi, Viet Nam
[g] Ho Chi Minh City Open University, Ho Chi Minh, Viet Nam
[h] Undergraduate Course of Environmental Engineering, Thuyloi University, 175 Tay Son, Dong Da, Hanoi, Viet Nam

ARTICLE INFO

Keywords:
Water quality index
Machine learning
Feature selection
Embedded method
Random Forest

ABSTRACT

A variety of water quality indices have been used to assess the state of waterbodies all over the world. In calculating a Water Quality Index (WQI), traditional methods require the evaluation of many water quality parameters, making them costly and time-consuming. In recent years, machine learning (ML) algorithms have emerged as an effective tool to solve many environmental problems, including water quality management. In this study, we investigate the performance of the ML-based method in calculating the WQI. We apply several feature selection techniques to select the key parameters fed the ML models. Experiments are carried out to evaluate the WQI based on a dataset collected from 2007 to 2020 of An Kim Hai system, one of the most important irrigation systems in the north of Vietnam. The obtained results show that the application of selection methods allows reducing significantly the number of water quality parameters fed the ML models without losing their accuracy. In particular, by using the embedded method, we find out four important parameters, including Coliform, DO, Turbidity, and TSS, that have the greatest impact on water quality. Based on these parameters, the Random Forest model provides the best accuracy in predicting the WQI values from the An Kim Hai system with a Similarity of 0.94. The combination of feature selection and ML methods is then considered an effective alternative for calculating the WQI, leading to a desirable performance and a reduction of input parameters. This makes water quality monitoring less costly, substantial effort, and time.

1. Introduction

Surface water is an important resource for the environment and for human life. Fresh surface water sustains ecological systems, provides a habitat for countless aquatic animals and plants, and supports many human uses like drinking water, domestic uses, irrigation, livestock, commercial uses, and industrial uses. Unfortunately, human exploitation has negative effects on water quality. Daily domestic wastewater, industrial wastewater, and agricultural wastewater have polluted surface water resources in rivers, lakes, ponds, and streams. In turn, the deterioration of surface water quality causes serious impacts on the surrounding ecosystem. In order to prevent those negative effects, monitoring and assessing the quality of surface water is an important task, allowing managers to take necessary actions to improve water quality once it is contaminated.

The surface water quality is determined by the physical, chemical, and biological components contained. These elements arise from natural processes and human activities. For example, the water flow erodes the banks, bringing with it mud, and leading to turbidity and suspended solids. Wastewater from livestock and farming activities contains biological ingredients like bacteria, viruses, and parasitic worms, and chemical ingredients like pH, phosphates, pesticides, and ammonical nitrogen. Meanwhile, industrial wastewater from manufacturing factories may carry heavy metals and surfactants. That is to say, there are

* Corresponding authors.
E-mail addresses: buiquoclap@tlu.edu.vn (B.Q. Lap), hongptt11@fe.edu.vn, hongptvn@gmail.com (T.-T.-H. Phan).

https://doi.org/10.1016/j.ecoinf.2023.101991
Received 12 June 2022; Received in revised form 10 January 2023; Accepted 10 January 2023
Available online 18 January 2023
1574-9541/© 2023 Published by Elsevier B.V.

many sources and factors that affect water quality. This makes the assessment of water quality more difficult. In this context, a multitude of Water Quality Indices (WQIs) have been introduced as an effective measure for the overall assessment of water quality. A WQI typically integrates many physical, chemical, and biological parameters into a single value that can be used to describe the general health or status of a waterbody. The value of WQIs typically ranges from 0 to 100, where the larger the WQI value is, the better the water quality is.

In the literature, there are several ways to calculate the WQI. Traditional methods formulate a single WQI as a combination of many sub-indices. That means, to calculate the WQI, these methods require measuring a large number of water quality parameters. As a consequence, they are often costly and time-consuming. This is a really big obstacle in assessing water quality, especially for developing countries, where infrastructure and investment are limited. Recently, machine learning (ML) based methods have been extensively used for calculating the WQI. Due to its power in handling complex nonlinear relational data, ML algorithms can mine potential patterns and discover the underlying mechanisms in the data, and then provide a good prediction for the WQI value.

When calculating the WQI, the first step of the ML-based method is to select the crucial water quality parameters. This selection is very important as it has a great impact on the effectiveness of the calculation of the WQI values. The main advantage of the ML-based method over the traditional methods is that they aim to select as few parameters as possible and achieve the most possible accuracy. From this point of view, one should consider that an ML-based model has no practical application if it uses the same input as the traditional methods. In the literature, most of the existing studies applied statistical methods such as Pearson correlation coefficient (Asadollah et al., 2021; Kocer and Sevgili, 2014), the PCA method (Kim et al., 2017; Jiang et al., 2020), or the expert opinions (Uddin et al., 2021) to determine important water quality parameters. Although these methods worked in some case studies, they do not take advantage of the other feature selection algorithms in machine learning. Some studies even tried all possible combinations of parameters to find the most effective ones. This certainly not only consumes time and computational effort but also makes them difficult to be applied in practice. To overcome these limitations, in this study, we propose a novel ML-based method that combines feature selection techniques with some ML algorithms for calculating the WQI. The performance of the proposed method is validated in a case study in which we evaluate the WQI based on a dataset collected from the AKH irrigation system, an important irrigation system in the North of Vietnam. The obtained results show that our proposed methods can significantly reduce the number of water quality parameters without losing the accuracy in calculating WQI. To the best of our knowledge, this is the first study in the literature exploiting different kinds of feature selection methods to automatically find the parameters that have the most influence on water quality, resulting in less use of parameters but keeping the performance of the ML models. The main contributions of this study are summarized as follows:

1)We propose a novel ML-based method that combines feature selection methods and machine learning algorithms to estimate the WQI values. Experiments on a real dataset show that our proposed method leads to a good prediction of WQI values with a similarity up to 0.94.
2)We explore the power of various feature selection techniques to find the most influential parameters included in the water. These techniques allow us to reduce significantly and efficiently the number of water quality parameters in predicting the WQI. In particular, by using the embedded method and the Random Forest algorithm, the number of parameters can be reduced from 10 to 4 while achieving the best accuracy of calculating WQI.
3)We prove through a case study that the ML-based method can be considered an effective alternative for calculating the WQI. These methods not only accurately evaluate the WQI values, but also save time and analytical costs considerably. This finding has a practical meaning, especially for the cases where it is not straightforward to measure water quality parameters.

The rest of the paper is organized as follows: In Section 2, we present a brief review of the related work in the literature. Section 3 provides information about the case study, including the study area, the data collection, and the traditional method to calculate the WQI values in Vietnam. The proposed ML-based method for computing the WQI has been presented in Section 3. Section 4 will explain the experiments and discuss the obtained results. Finally, some concluding remarks are given in Section 5.

2. Related work

The WQI concept has a long history and has been widely used worldwide. In the US, the index was developed under the support of the National Sanitation Foundation, called the NSF-WQI (Brown et al., 1970). The Canadian Council of Ministers of the Environment (CCME) provided the CCME WQI for water quality monitoring and assessment of surface and underground water in terms of their chemical, biological and nutrient constituents and overall aesthetic condition (Miller et al., 1986). The Scottish WQI, introduced by the Engineering Division of the Scottish Research Development Department, was also used in Spain, Portugal, Thailand, and Iran (Sutadian et al., 2016). Malaysia selected 6 parameters to determine WQI (Shuhaimi-Othman et al., 2007; Najah et al., 2011). Indonesia developed a tool, called the West Java Water Quality Index, to assess water quality in rivers of the West Java Province (Sutadian et al., 2018). Uddin et al. (2021) conducted a comprehensive review of 21 among more than 35 WQI models introduced by various countries or agencies and their use for assessing surface water quality around the world.

To calculate the WQI, several methods have been suggested in the literature. For instance, Pesce and Wunderlin (2000) proposed a general formula for the WQI that includes the normalized values of each parameter, and the corresponding weights. This formula is consistent with the one applied in many other studies (Misaghi et al., 2017; Fathi et al., 2018; Wu et al., 2018, 2021). A summary of various WQI formulas, including four weighted and three non-weighted ones, was presented in Uddin et al. (2021). The sub-indexing functions and aggregating functions for these formulas were further discussed in Uddin et al. (2022). Another critical review of the WQI formulas and the corresponding development can be found in Gupta and Gupta (2021); Rezaie-Balf et al. (2020).

As discussed in Bui et al. (2020), manual WQI calculations have certain disadvantages such as requiring the measurement of many parameters, being inconsistent due to the use of different equations, and being fraught with errors during derivations of sub-indices. To overcome these issues, the ML-based method has been proposed and applied successfully for predicting the WQI in the literature. The use of these methods is becoming increasingly popular in recent times and the references are abundant. Hameed et al. (2017) developed a back-propagation neural network (BPNN) and radial basis function neural network (RBFNN) to determine the relationship between WQI and chemical parameters included in tropical environments. An artificial neural network model was built in Othman et al. (2020) to deal with the problem of missing some water quality parameters for the input required by the manual method for calculating the WQI. A novel ML approach for estimating and predicting uncertainty in a WQI model was presented in Uddin et al. (2022). Asadollah et al. (2021) investigated the performance of the three ML algorithms, including the Decision Tree (DT), Extra Tree (DT), and Suppor Vector Machine (SVM), in calculating the WQI of the Lam Tsuen River in Hong Kong. In Ho et al. (2019), the DT technique was used to predict the WQI for the Klang River in Malaysia and its classification within a specific water quality class. Many other

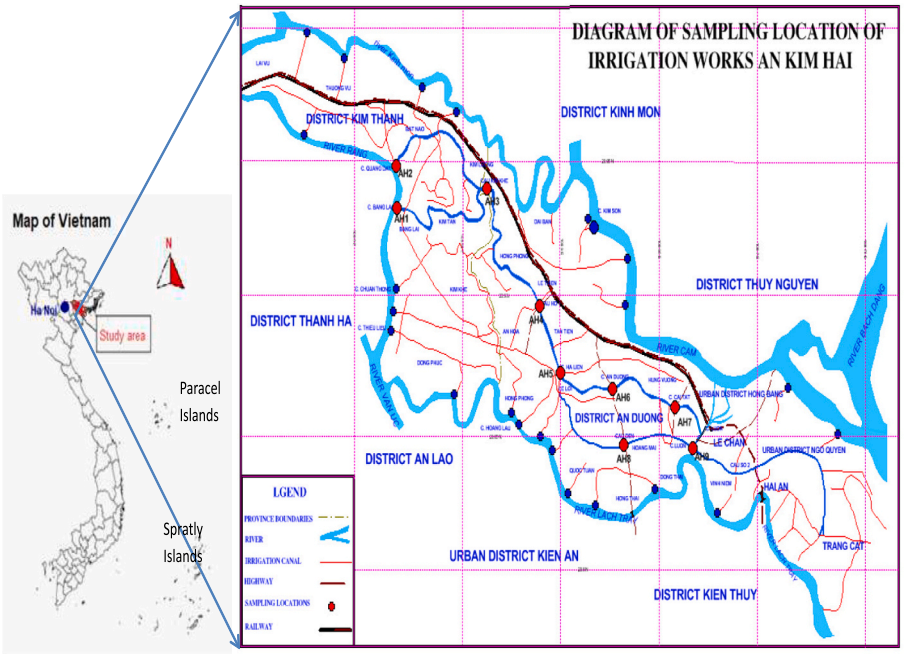


Fig. 1. An Kim Hai irrigation system and sampling locations.

authors also applied several ML algorithms at the same time for a dataset to compare their performance and find the model that can most accurately predict the WQI value, see, for example, Chou et al. (2018), Derdour et al. (2022), Suwadi et al. (2022), Shamsuddin et al. (2022), Shamsuddin et al. (2022) and Manzar et al. (2022).

In the use of ML models, feature selection is to determine a subset of input variables that can accurately represent data while minimizing the impact of irrelevant factors and still achieving precise results. Most of the existing studies in the literature have utilized a number of statistical approaches for selecting the key water quality parameters. Some authors applied the "trial and error" method to find the best combination of parameters fitted to their ML models (Bui et al., 2020; Ahmed et al., 2019; Khoi et al., 2022). These methods are quite cumbersome with complex calculations and may not always be effective (Uddin et al., 2022). Very recently, Uddin et al. (2022) used the extreme gradient boosting (XGBoost) algorithm to rank water indicators based on their measured values and influence on water quality status.

3. The study area and materials

3.1. The An Kim Hai irrigation system

An Kim Hai (AKH) is an irrigation system in the Northern Coastal Plain in Vietnam, sandwiched between the right dike of the Cam River and the left dike of the Lach Tray River (Fig. 1). This is an inter-provincial system of irrigation between two big provinces in the North of Vietnam, Hai Phong City, and Hai Duong Province. The total natural area of the AKH irrigation system is about 36,570 hectares. It is responsible for irrigating 15,946 hectares of cultivation, most of which are for rice (the spring season from January to March and the summer season from May to July) and vegetables (the winter season from November to December). The system also plays a role in water supply and drainage for residential areas, craft villages, and livestock areas in two provinces. For example, it supplies the input to water plants with an output of over 55 million m^3/year, equivalent to about 80% of the domestic water of Hai Phong City. As a result, AKH Irrigation System has to receive a large amount of wastewater from domestic, agricultural, industrial, and other activities. Monitoring and evaluating the water quality of the system is therefore an important mission, contributing significantly to the sustainable development as well as environmental protection of the region.

3.2. The data collection

Like other cases, the Vietnam water quality index (VN_WQI) is calculated based on many parameters representing the water quality condition. These parameters have been split into five groups according to a technical guideline for calculation and publication of the VN_WQI (MONRE, 2019). The most popular used parameters include pH, Turbidity (Turb), Temperature, Total Suspended Solids (TSS), Dissolve

Table 1
Descriptive statistics of water quality parameters.

S.N	Variables	Acronyms	Units	Min	Max	Median	Mean	STD
1	pH	pH	pH unit	5.1	7.9	6.9	6.8	0.4
2	Turbidity	Turb	NTU	5.4	642	96	114.6	67.5
3	Total Suspended Solids	TSS	mg/L	2.3	1247.8	97	110.4	116.8
4	Ammoniacal nitrogen	NH_4^+ −N	mg/L	0.004	37	0.9	2.2	4.7
5	Phosphate	PO_4^{3-} −P	mg/L	0.01	5.5	0.1	0.5	0.9
6	Dissolved oxygen	DO	mg/L	1.5	8.5	6.2	5.3	1.9
7	Chemical Oxygen Demand	COD	mg/L	4.8	580	20.7	29.3	35.4
8	Biochemical Oxygen Demand	BOD_5	mg/L	2.0	130.6	9.6	13.2	12.2
9	Coliform	TC	MPN	45	5400000	5100	50003	332220
10	Temprature	T	°C	24.5	36.0	27.5	27.8	1.1

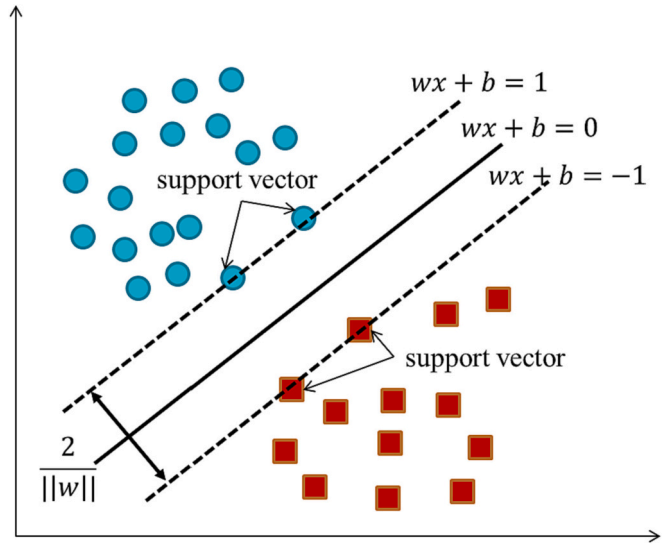


Fig. 2. Linear classification with SVM.

Oxygen (DO), Biological Oxygen Demand (BOD_5), Nitrogen (NH_4^+ −N), Phosphates (PO_4^{3-} −P), Chemical Oxygen Demand (COD), and Total Coliform (TC) are shown in Table 1.

Following the guideline, a dataset related to these 10 parameters has been collected at different locations in the AKH irrigation system. Fig. 1 provides a picture of these locations in the system. A total of 657 samples in the dataset have been collected during the months of the dry and rainy seasons in 12 consecutive years (from 2007 to 2020). Each sample comprises the values of these 10 water quality parameters. In the following sections, we call the collected dataset the AKH dataset.

3.3. The traditional method to calculate the WQI Vietnam

In the technical guideline, the value of each water quality parameter is denoted by WQI_{SI} (SI stands for sub-index). Once the WQI_{SI} is calculated for all the involved parameters, the final VN_WQI is defined by the formula:

$$WQI = \frac{WQI_{PH}}{100} \times \left[\frac{1}{5}\sum_{i=1}^{5} WQI_{a_i} \times \frac{1}{2}\sum_{i=1}^{2} WQI_{bj} \times WQI_c\right]^{\frac{1}{3}} \quad (1)$$

where WQI_{PH} stands for the value of pH index, WQI_c is for the value of Total Coliform, $WQI_{a_i}, i = 1, \ldots, 5$ represents the values of five indexes: DO, BOD_5, COD, NH_4^+ −N, and PO_4^{3-} −P, and $WQI_{b_j}, j = 1, 2$ are the values of TSS and Turbidity. Based on the obtained VN_WQI value, the quality of water is classified and the suitability for use of the water is stated based on the scale presented in Table 7.

In this study, the values achieved from this manual formula are considered as the *true* WQI values and they will be compared with the predicted WQI values to evaluate the performance of the ML models.

4. The proposed ML methodology for calculating the WQI

In this section, we present the proposed ML-based method to calculate the WQI. Each of the algorithms applied in the study is described briefly in the following.

4.1. The ML models

4.1.1. Multi-linear Regression

Multi-linear Regression (MLR) is a type of linear regression that extracts the linear relationship between the target variable and one or more predictor variables. This method evaluates the input of each variable that affects the dependent one, as expressed in the equation (Amral et al., 2007):

$$y = \alpha_0 + \alpha_1 x_1 + \alpha_2 x_2 + \ldots + \alpha_k x_k + \epsilon, \quad (2)$$

where x_i is the value of the ith predictor, α_i is the coefficient of the ith predictor, α_0 is the regression constant and ϵ is the error term. MLR fits the model using least squares. This means that the sum of differences squares between the observed and predicted values is minimized.

4.1.2. Support Vector Machine

Support Vector Machine (SVM) is a statistical machine learning algorithm developed by Vapnik (1995) for solving high-dimensional and nonlinear pattern recognition. The main idea of this method is to find

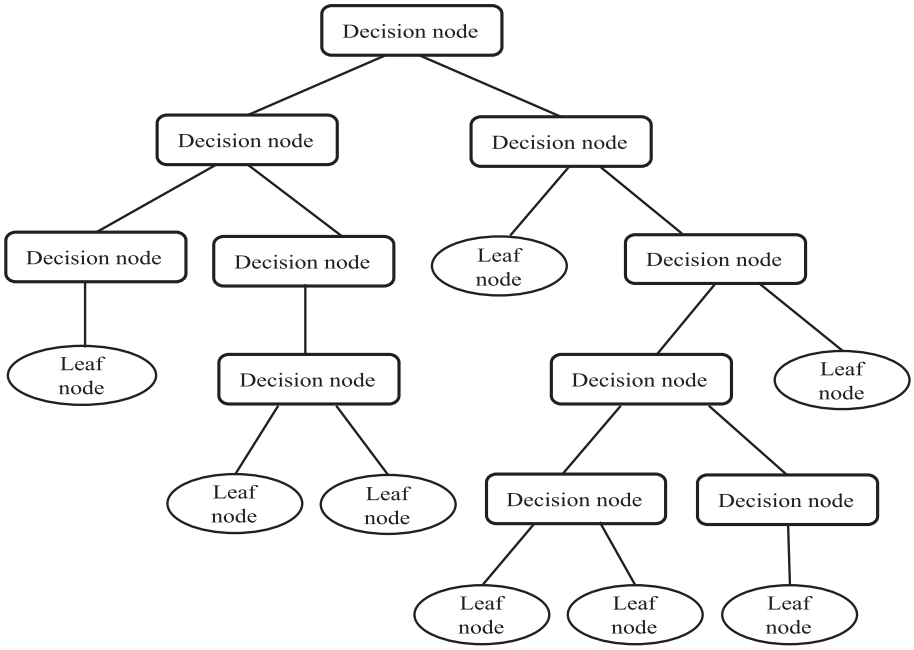


Fig. 3. A diagram of the Decision Tree algorithm.

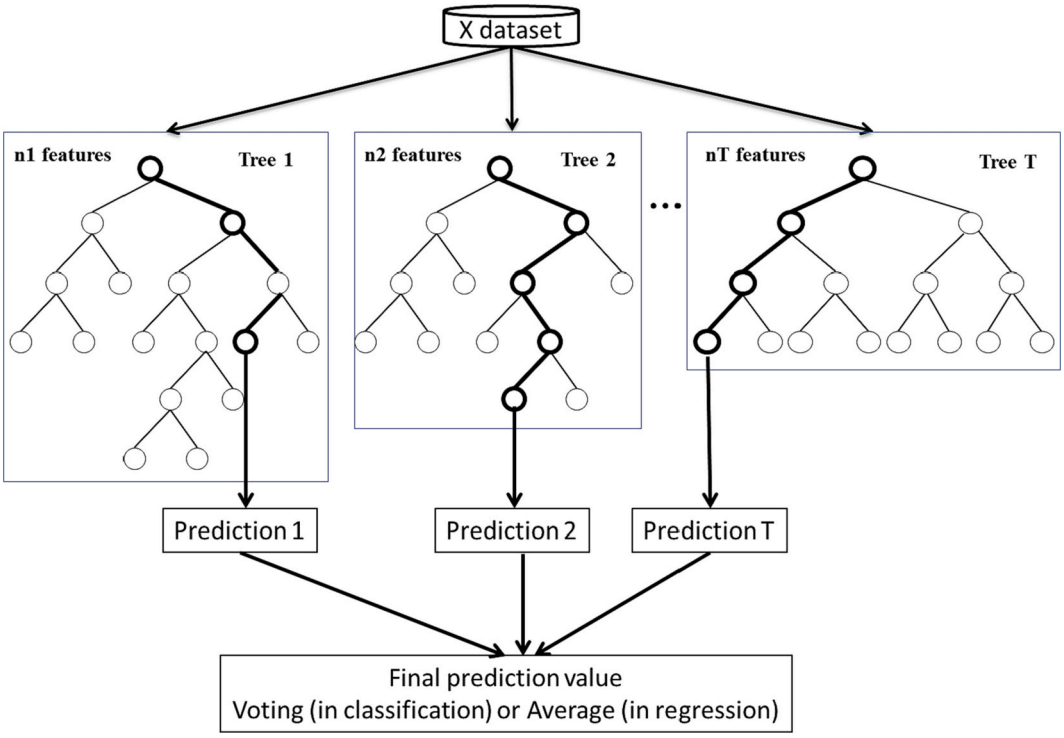


Fig. 4. A diagram of the process of the Random Forest algorithm.

the best hyper-plane for separating linear patterns in a high dimensional space where features are mapped onto. There are many hyperplanes satisfying this criterion. So SVM will discover the optimal hyper-plane having the largest margin between the hyper-plane and the nearest input points (Fig. 2). This is done with the help of the data points that are located closest to the decision surface and have direct with the optimum location of the decision surface.

SVM can deal with nonlinear classification or regression problems by transforming original data into a new space using kernel functions. The new space, called feature space, is usually highly dimensional so that the classes become linearly separable.

4.1.3. Decision Tree

A Decision Tree (DT) is a popular machine-learning technique that can be applied to both classification and regression problems. The objective is to build a model that predicts the value of a dependent variable by learning simple decision rules inferred from the features of the data. Fig. 3 illustrates the idea of this technique. A DT is a structured hierarchical tree used to classify objects based on a series of rules. Giving objects with features and their classes, the decision tree will generate rules to predict the class of unseen data. The training set is used to build the decision tree. The algorithm divides the data set into 2 branches based on the binary split process at each node. This separation procedure is then applied to develop new branches until each branch becomes inseparable.

4.1.4. Random Forest

Random Forests (RF) is an ensemble learning method used for solving both regression and classification problems. This algorithm creates multiple dissimilar trees from random subspace sampling of the training data. It builds fully grown trees without pruning and selects a random subset of attributes at each node to split to keep the low bias. The randomness in the process of building trees in the forest increases the diversity and reduces the correlation between trees. This makes RF more accurate and stable when adding more trees to the forest. Fig. 4 demonstrates the idea of a random forest algorithm.

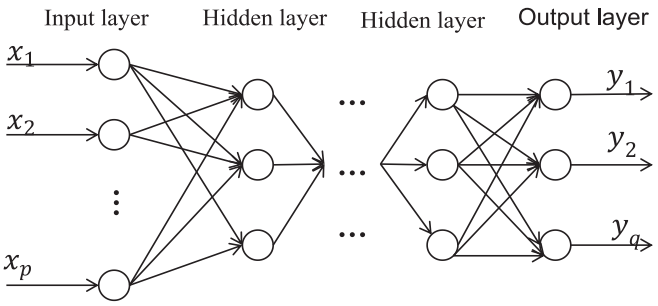


Fig. 5. Architecture of a general MLP network.

When building a tree, an additional tuning parameter is the number of candidate variables selected for node splitting at each iteration, called *mtry*, and $1 < mtry < n$. An objective of selecting $mtry < n$ is to reduce the computational time. Normally, *mtry* is chosen as $mtry = \sqrt{n}$ for classification problems and $mtry = n/3$ for regression ones. Another parameter to consider is *ntree*. Trees usually tend to be unstable with high variance. In the random forest algorithm, a number of trees (*ntree*) is used to reduce the variance.

4.1.5. Multi-layer Perceptron

Multi-layer Perceptron (MLP) model is a class of feed-forward neural networks that was inspired by the interconnection neurons of the human. The architecture of a general MLP network can be described in Fig. 5. It consists of at least three types of layers: the input layer, output layer, and hidden layer in which the data flow in the forward direction from the input to the output layer. Every neuron in the next layer is connected to all the neurons in the previous layer. The output of the previous layer is the input of the next layer neuron. The hidden layers between the input and output layers are the real computational engine of MLP. The recognition/prediction/classification task is performed at the output layer. In the MLP, back-propagation is used as a supervised learning technique.

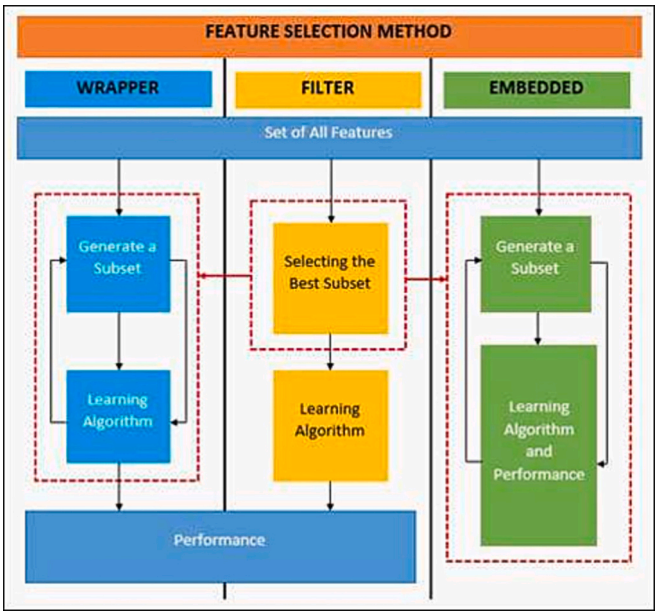


Fig. 6. Category of supervised feature selection and workflow of each method (Rosely et al., 2019).

The input layer gets the input vector of p-dimension $(x_1, x_2, \ldots, x_p)$ for processing. The input data are passed to the first hidden layer. At this layer, the data are processed and then passed through the transfer function to produce the result. This result, as the input signal, will be transmitted to the neurons of the next hidden layer. The process continues until the neurons at the output layer give final results - a vector of q-dimension $(y_1, y_2, \ldots, y_q)$.

4.2. Feature selection methods

Feature selection refers to a technique of selecting key features from a dataset. It aims to reduce the number of input variables by removing redundant or irrelevant features and choosing the most relevant subset of parameters to the ML models. Several benefits of applying feature selection methods have been discussed in the literature such as reducing variance, shortening training time, and increasing the model accuracy.

These methods are divided into two main categories, supervised and unsupervised (Song et al., 2007). The supervised types are further classified into filter models, wrapper models, and embedded models as shown in Fig. 6 and will be applied in our study.

- *Filter methods:* The Filter methods calculate a score for each feature based on uni-variate statistical measures like correlation coefficients and Fisher's score, and then filter the features with the highest scores. One of the advantages of this kind of method is that they are independent of the learning algorithms and take less computational time than other methods. An overview of existing filter methods can be seen in Bommert et al. (2020)
- *Wrapper methods:* The wrapper methods try different subsets of features based on several predefined learning models and those features that result in the best-performing model according to a performance metric are selected. Due to the repetition of the learning phase and cross-validation, these methods are more computationally expensive than filter methods in general. The main idea of a wrapper model involves the following steps:
  - –Step 1: Look for a subset of features
  - –Step 2: Feed to a machine learning method and evaluate the selected subsets based on the classifier performance
  - –Step 3: Repeat Step 1 and Step 2 until the desired result is achieved

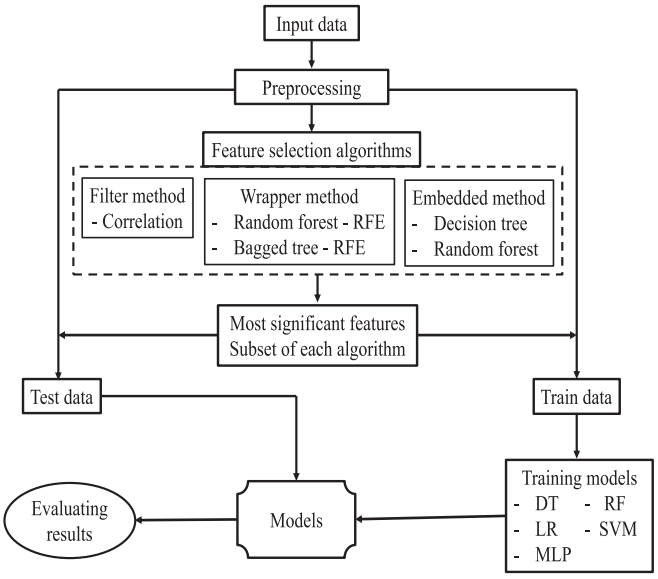


Fig. 7. A visualization of the proposed ML-based method for the WQI prediction.

In the first step, the greedy search strategy is widely used. It includes two flavors: forward selection and backward elimination. While forward selection progressively incorporates variables into larger subsets, backward elimination starts with the set of all variables and then progressively eliminates the least promising ones. Recursive feature elimination (RFE) is one of the most popular backward elimination methods that is commonly used together with some classification ML algorithms (Yan and Zhang, 2015).

- *Embedded methods*: The methods take advantage of both filter methods and wrapper methods. An embedded algorithm uses statistical criteria to select several candidate feature subsets and then chooses the subset with the highest performance during the training process. That is, it incorporates the feature selection as part of the model construction process, and model fitting and feature selection are performed simultaneously. The most popular embedded technique is the tree-based algorithms like RF, DT, etc. (Mitra et al., 2002).

4.3. The proposed architecture of the ML-based models for the WQI prediction

Fig. 7 provides a visualization of our proposed method for predicting the WQI value. The objective of the proposal is to take into account the advantages of feature selection algorithms to select key water quality parameters before feeding them to ML models. The architecture of our proposed method involves three stages: i) pre-processing data, (ii) selecting features, and (iii) predicting the WQI.

In the first stage, the raw data was collected and processed by removing noise (the data points that are out of a standard range), correcting the data inconsistencies, and integrating them into a homogeneous dataset. After pre-processing, the critical water quality parameters are selected based on the filter methods, the wrapper methods, and the embedded methods. We propose conducting several different feature selection methods to find the most important set of parameters. Then, the data from these parameters are split into training and testing sets. The training data will be used to train the ML models, namely Decision Tree, Multi-linear Regression, Multi-layer Perception, Random Forest, and Support Vector Machine. The performance in predicting the WQI of these models is validated on the testing set.

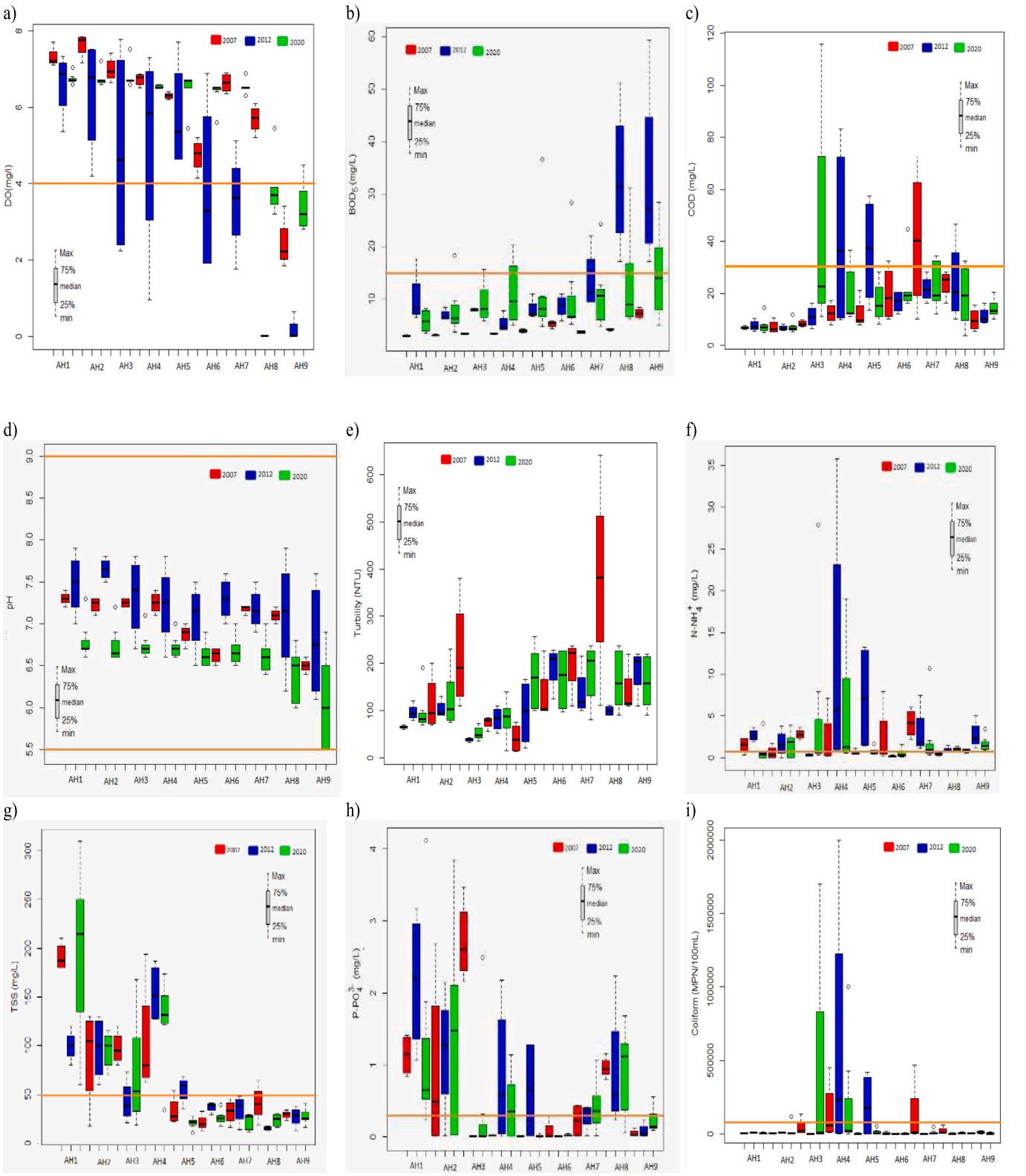


Fig. 8. Spatial and temporal variation of the water quality parameters in 3 representative years (2007, 2012 & 2020).

4.4. Metrics to evaluate the performance of ML models

We evaluate the performance of the ML methods by comparing the predicted WQI with the *true* WQI. The comparison is conducted based on seven popular metrics, including Similarity (Sim), *R* score, the Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Fraction of Standard Deviation (FSD), FA2 and Nash Sutcliffe efficiency (NSE). These metrics are possessed important properties to assess effectively the performance of predictive ML models from different perspectives. The metrics are briefly described as follows.

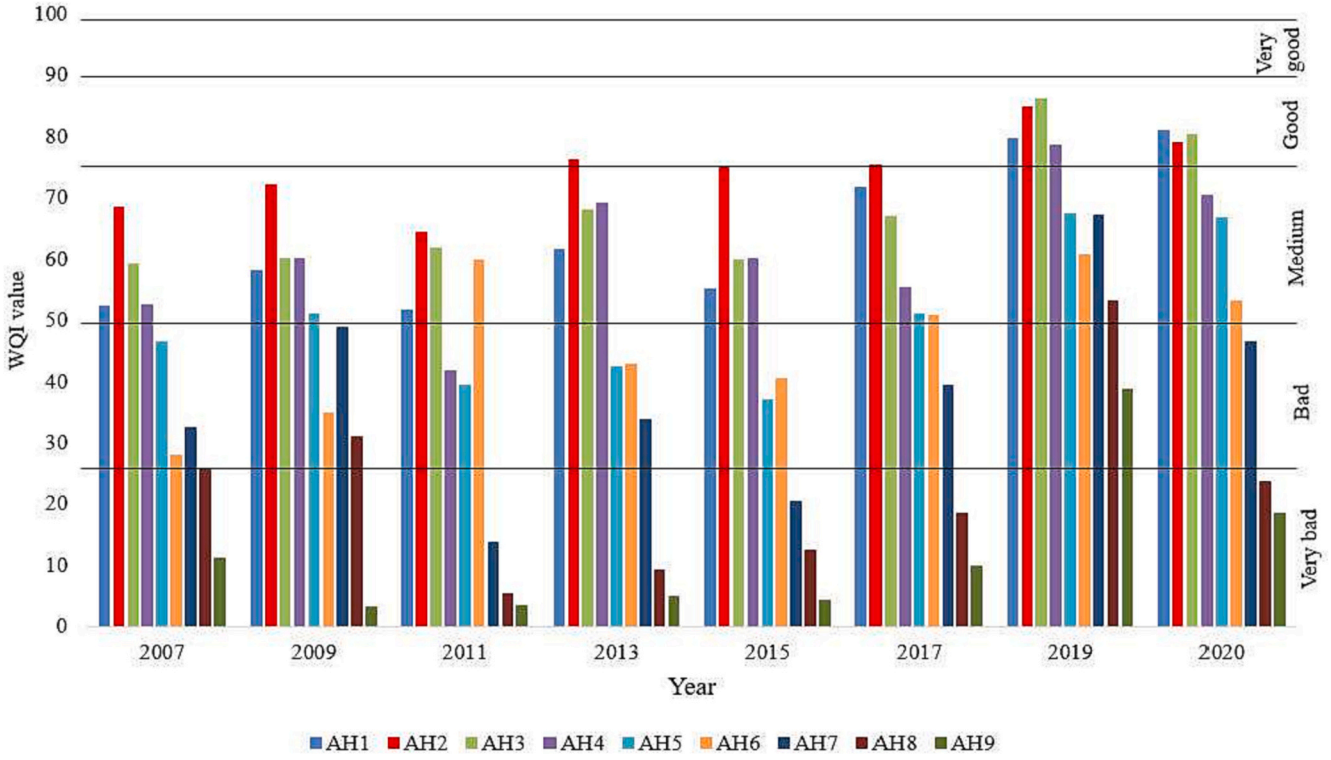


Fig. 9. Average WQI value from 2007 to 2020.

- Similarity: The metric determines the percentage of similar values between the predicted values, $\mathbf{y} = (y_1, \ldots, y_m)$, and the true values, $\mathbf{x} = (x_1, \ldots, x_m)$. It is defined by the formula

$$Sim\left(\mathbf{y}, \mathbf{x}\right) = \frac{1}{m}\sum_{i=1}^{m}\frac{1}{1+\frac{|y_i - x_i|}{\max(x)-\min(x)}} \tag{3}$$

where m is the number of predicted values. The value of Similarity is in the $[0, 1]$ interval. A higher similarity shows a better ability of the method for the prediction task.

- R score: The score is determined as the correlation coefficient between two variables $\mathbf{y}$ and $\mathbf{x}$. The R score is also between 0 and 1. A method presents better performance when its R score is higher.
- RMSE: The Root Mean Square Error is computed as the average squared difference between the forecast values $\mathbf{y}$ and the respective true values $\mathbf{x}$. This indicator is very useful for measuring the overall precision or accuracy of a method. In general, the most effective method would have the lowest RMSE, where

$$RMSE\left(\mathbf{y}, \mathbf{x}\right) = \sqrt{\frac{1}{m}\sum_{i=1}^{m}(y_i - x_i)^2}. \tag{4}$$

- MAE: The Mean Absolute Error between the predicted values $\mathbf{y}$ and the actual ones, $\mathbf{x}$, is defined as:

$$MAE\left(\mathbf{y}, \mathbf{x}\right) = \frac{1}{m}\sum_{i=1}^{m}|y_i - x_i| \tag{5}$$

The lower the MAE value, the better the performance method.

- FSD: The Fraction of Standard Deviation points out whether a method is acceptable or not. the fraction is determined by

$$FSD\left(\mathbf{y}, \mathbf{x}\right) = 2*\frac{|SD(y) - SD(x)|}{SD(y) + SD(x)} \tag{6}$$

The closer the value of FSD to 0, the more impeccable the method.

- FA2: The FA2 represents the fraction of data points that satisfied the smoothing amplitude cover. A model is considered to be better when its FA2 is closer to 1.

$$FA2\left(\mathbf{y}, \mathbf{x}\right) = \frac{length\left(0.5 \leqslant \frac{y}{x} \leqslant 2\right)}{length(x)} \tag{7}$$

- NSE: The Nash Sutcliffe efficiency is used to evaluate the predictive ability of hydrological models. The NSE values range from -∞ to 1, where higher values mean a better fit between observed and forecast water level (Nash and Sutcliffe, 1970).

$$NSE\left(\mathbf{y}, \mathbf{x}\right) = 1 - \frac{\sum_{i=1}^{m}(x_i - y_i)^2}{\sum_{i=1}^{m}(x_i - \overline{x_i})^2} \tag{8}$$

5. Experiments and results

In this section, we first analyze spatial and temporal water quality characteristics of the AKH system. Then, we present the experiments based on the proposed method and discuss the obtained results. In our experiments, the AKH dataset is split into two subsets, namely the training set (accounting for 67% of samples) and the testing set (taking the rest of 33% of samples). The training set is used to train the ML models while the testing set is to test the efficiency of the trained models. We consider two scenarios. In the first scenario, we apply all the parameters from the dataset to explore the performance of the ML

methods. Once the efficiency of these models is validated, the second scenario is executed, aiming to find out a way to reduce the number of water quality parameters fed to the ML models.

5.1. Spatial and temporal variation of the water quality parameters

We first focus on the spatial and temporal variation of the water quality parameters of the AKH irrigation system for only three representative years of the monitored period. Fig. 8 presents the variation of each parameter for the years 2007 (the first period), 2012 (the middle period), and 2020 (the last period). The horizontal line in each sub-figure of Fig. 8 indicates the criteria threshold corresponding to each parameter according to the guideline in Vietnam.

As can be seen from this figure, the monitored water quality parameters fluctuate significantly both spatially and temporally in general. In particular, among three representative years of monitoring, most of the median values of DO (sub-figure (a)) were higher than the allowable threshold (4 mg/L) for the irrigation water quality criteria in Vietnam, except for station AH9. The same results were also observed for PO_4^{3-} −P (sub-figure (h)) and NH_4^+ −N (sub-figure (f)). In more detail, the highest median values of PO_4^{3-} −P were seen at the locations AH1 and AH2, while the highest median values of NH_4^+ −N were in 2012, especially at the stations AH4 and AH5. There were some stations where the NH_4^+ −N median value was slightly below the threshold like AH3 (2012), AH5 (2007), AH6, and AH8 (in all three years). In contrast, most of the median values of BOD_5 (sub-figure (b)) and COD (sub-figure (c)) were lower than the allowable threshold of 15 mg/L and 30 mg/L, respectively. This was also true for Coliform (sub-figure (i)), thereby indicating that the water quality was safe from pathogens and can be used for irrigation purposes. Meanwhile, the fluctuation range of pH (sub-figure (d)) was not large over the years: all of its values were within the permissible range from 5.5 to 9.0. Moreover, the pH median values showed a decreasing tendency in three representative years. Finally, the TSS (sub-figure (g)) median values were much higher than the permissible threshold of 50 mg/L) for the stations from AH1 to AH4, but were acceptable for irrigation for the stations from AH5 to AH9. There are no standard limits of Turbidity stated for the quality of irrigation water in Vietnam.

When it comes to the entire monitoring period from 2007 to 2020, the statistical analysis of the parameters is summarized in Table 1 while Fig. 9 presents the annual average VN_WQI values.

In general, the VN_WQI showed a decreasing tendency from AH1 (upstream) to AH9 (downstream) for every year of the monitoring period. This is a reasonable observation due to the fact that pollutants usually flow from upstream to downstream regions, leading to a better water quality upstream compared to that downstream. In addition, the VN_WQI values appeared to increase slightly from 2007 to 2020. That could be the result of measures from the local authorities that improved water quality in the AKH irrigation system year by year. In particular, the average VN_WQI at station AH9 (the end of downstream) was always in the "very bad" range (less than 25) for every year from 2007 to 2020. From station AH1 to AH4, this value was in the "medium" range (from 50 to 75) for most of the years monitored. Some stations even experienced a remarkable change in the last years (2018–2020), changing from "medium" to "good" quality with WQI values greater than 75. At the same time, the VN_WQI values were mostly in the "bad" range (25–50) at stations AH5, AH6, and AH7 and fluctuated from the "very bad" to "bad" range at station AH8. Fig. 9 also indicated that the water quality in the "medium" range accounted for 40.5%, in the "bad" condition accounted for 27 %, in the "very bad" condition was 20.6%, and in the "good" condition was 11.9%. There was no "very good" condition of the water quality throughout the monitoring period.

After initial analysis of spatial and temporal water quality evolution, we continuously conduct experiments on the AKH dataset using the proposed method. Detailed results and discussion are demonstrated in the next section.

Table 2
The performance of ML modes based on all features (best results in bold).

Method	Sim	R	RMSE	MAE	FSD	FA2	NSE
RF	**0.934**	**0.92**	**11.03**	**6.78**	0.16	**0.93**	**0.77**
DT	0.921	0.89	12.60	8.99	**0.11**	0.85	0.74
MLP	0.902	0.86	14.17	11.16	0.27	0.78	0.53
SVM	0.880	0.72	18.82	14.44	0.32	0.73	0.09
LR	0.856	0.65	20.82	17.40	0.52	0.65	−0.71

Table 3
The performance of ML models based on the features selected using the Correlation filter method (best results are in bold).

Method	Sim	R	RMSE	MAE	FSD	FA2	NSE
RF	**0.932**	**0.92**	**11.03**	**7.03**	0.18	**0.91**	**0.77**
DT	0.921	0.89	12.58	8.99	**0.11**	0.85	0.73
MLP	0.904	0.86	13.98	10.97	0.25	0.81	0.57
SVM	0.882	0.73	18.66	14.21	0.34	0.76	0.07
LR	0.857	0.65	20.71	17.25	0.51	0.67	−0.63

5.2. The performance of ML methods based on all features

In this scheme, the input data corresponding to 10 parameters, including pH, Temperature, Turbidity, TSS, BOD_5, COD, DO, PO_4^{3-} −P, NH_4^+ −N, and Coliform, are used to train the ML models. After training, these models are used to compute the WQI with the input from the testing set. Then, we compare the predicted WQI values and the *true* WQI values using the seven metrics mentioned above. The obtained results are presented in Table 2. In general, the ML models, especially the tree-based models, show a good performance in calculating the WQI. The RF algorithm produces the best performance with the highest values of Similarity (0.934), R score (0.92), FA2 (0.93), and NSE (0.77), and the smallest values of RMSE (11.03) and MAE (6.78). The DT algorithm leads to the smallest value of FSD, while other metrics are not as good as the ones of the RF algorithm. For example, the Similarity between the WQI values predicted by the DT and the true WQIs is 0.921, slightly less than the case of using the RF algorithm. Other models like MLP, SVM, and LR are less efficient than these two tree-based algorithms. Table 2 also shows the consistency of the applied metrics. For example, using the RF algorithm leads to the largest values of similarity, R, FA2, and NSE (the larger the values of these indicators, the better the methods) and the smallest values of RMSE and MEA (the smaller the values of these indicators, the better the methods). By contrast, the LR algorithm leads to the smallest values of similarity, R, FA2, FSD, and NSE, and the largest values of RMSE and MAE.

The impressive efficiency of RF and DT algorithms in this experiment indicates that the ML methods could be considered an effective alternative for calculating the WQI. In the next experiments, we would like to explore whether the number of water quality parameters fed to the ML algorithms can be reduced without losing their performance.

5.3. The performance of ML methods based on the selected features

5.3.1. Using the filter method

In this experiment, we use the correlation coefficient metric to remove duplicate features. From a statistical point of view, a feature could be ignored when its correlation to others is large enough. According to a discussion in Ratner (2009), a cutoff value of $r = 0.7$ could be used to determine the variables that have a large correlation. By using this threshold and after calculating the 10×10 matrix of the correlation coefficients between 10 variables, we found that only two parameters BOD_5 and COD have a correlation coefficient greater than 0.7. Between these two features, we decided to remove the BOD_5 since it is

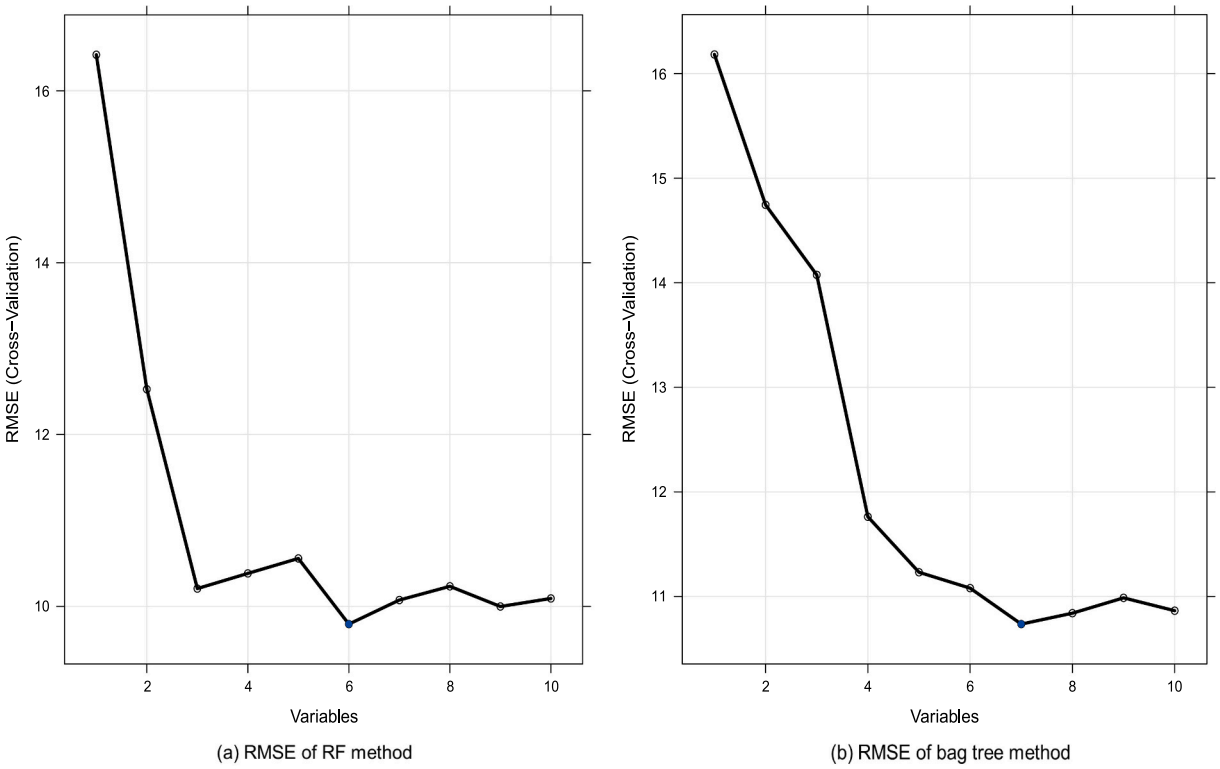


Fig. 10. RMSE of RF-RFE and BT-RFE algorithms.

corresponding to the largest mean absolute correlation.

After removing BOD_5, we perform exactly the steps as the previous experiment, but with only nine parameters. The values of metrics measured the difference between the predicted WQI and the *true* WQI are displayed in Table 3. As can be seen from this table, the values of these metrics have changed compared to the previous experiment with all 10 parameters, however, the change is insignificant. For example, for the RF model, the values of RMSE, R score, and NSE are still the same, the Sim and FA2 values decrease from 0.934 and 0.93 to 0.932 and 0.91, while the values of MAE and FSD increase from 6.78 and 0.16 to 7.08 and 0.18, respectively. The same observations are also witnessed for other metrics. That is to say, the ML algorithms remain their performance regardless of the elimination of BOD_5. That motivates us to conduct another experiment to explore if the number of parameters can be reduced more.

5.3.2. *Using the wrapper methods*

Compared to the filter method, the wrapper methods have certain advantages through interacting with the classifier for feature selection, considering feature dependencies, and better generalization. They also perform a more comprehensive search of feature space. Therefore, we use this technique to find key features from water parameters. In the use of wrapper methods, it is important to define the number of features to select. Choosing the right number of features that best represent the dataset has a significant impact on the performance of the ML models. A widely used method to choose an optimal value for this number relies on the RMSE metric. The smaller the RMSE value, the better the number of features selected. Due to the superior performance of the tree-based algorithms, in this study, we apply the RFE method based on the RF algorithm (denoted by RF-RFE) and the Bag trees algorithm (denoted by BT-RFE). Fig. 10 depicts the values of RMSE produced by these two algorithms for the different numbers of features of the AKH dataset. The results from the figure point out that the minimum value of RMSE is attained when using six parameters for the RF-RFE algorithm and seven parameters for the BT-RFE algorithm.

The next step in the experiment is to train the ML models based on the observations of these parameters from the AKH dataset. Then, similar to previous experiments, the trained models are applied to predict the WQI. Table 4 shows the difference between the anticipated WQI values and the *true* WQI values for both cases of using RF-RFE and BT-

Table 4
The performance of ML models based on selected RFE features (best results are in bold).

Selection method	ML method	Sim	R	RMSE	MAE	FSD	FA2	NSE
RF-RFE	RF	**0.933**	**0.9**	**11.08**	**6.57**	**0.08**	**0.94**	**0.79**
	DT	0.914	0.88	12.45	8.46	0.09	0.88	0.73
	SVM	0.848	0.6	21.32	16.44	0.24	0.66	−0.08
	MLP	0.846	0.6	21.1	16.61	0.31	0.66	−0.22
	LR	0.836	0.57	21.37	17.6	0.44	0.69	−0.66
BT-RFE	RF	**0.93**	**0.9**	**11.2**	**6.84**	**0.08**	**0.93**	**0.78**
	DT	0.851	0.65	19.85	15.72	0.33	0.71	−0.13
	SVM	0.848	0.6	21.32	16.44	0.24	0.66	−0.08
	MLP	0.846	0.6	21.1	16.61	0.31	0.66	−0.22
	LR	0.836	0.57	21.37	17.6	0.44	0.69	−0.66

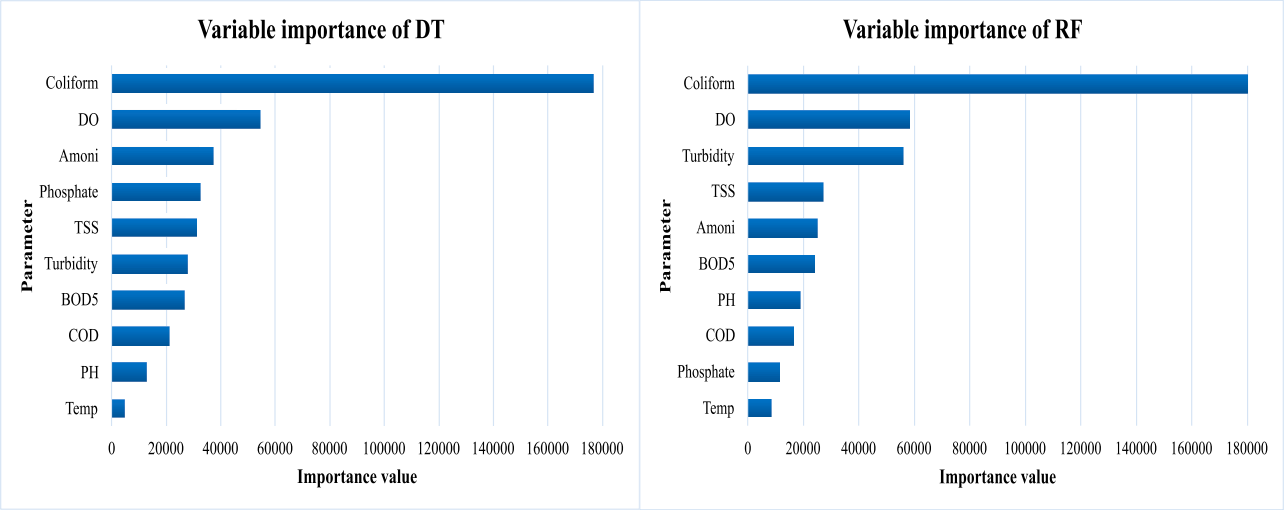


Fig. 11. Variable importance of RF and DT method.

RFE parameters. Some important conclusions can be drawn from this table as follow.

- RF is still the best model compared to others: it provides the best values concerning all the metrics evaluated. For instance, based on the RF-RFE features, it reaches the highest value for similarity (0.933), R score (0.9), FA2 (0.94), NSE (0.79) and the lowest value for RMSE (11.08), MAE (6.57), and FSD (0.08).
- The use of the RFE selection method allows a reduction from three to four water quality parameters while maintaining the accuracy of the ML models. Indeed, by using the selected RFE parameters, the RF and the DT models produce a similar performance compared to the case of using all the parameters. For example, based on six RF-FRE parameters, the RF model leads to an RMSE of 11.08, equivalent to 11.03, the one based on all ten parameters. For the DT model, this value is 12.45 compared to 12.60.
- The RF-RFE seems to be more efficient than the BT-RFE in this case study. Firstly, it requires fewer features (six compared to seven). Secondly, using the RF-RFE features leads to better performance for the RF and DT models. For example, the similarity of the two models based on the RF-RFE features are 0.933 (RF model) and 0.914 (DT model), and based on the BT-RFE features are 0.930 (RF model) and 0.851(DT model).

It has been pointed out in the literature that the embedded methods combine the qualities of both filter methods and wrapper methods: they are faster and more accurate than the filter methods while considering the interaction of features like in the wrapper methods. In the next section, the experiments with the embedded selection method will be discussed.

5.3.3. *Using the embedded methods*

As discussed in Mitra et al. (2002), the most popular embedded techniques are based on tree-based algorithms. Therefore, in this experiment, we apply the RF algorithm and the DT algorithm to select the water quality parameters. The parameters selected from these two algorithms will be called the RF features and the DT features.

In the process of building a tree, it is necessary to evaluate the variable importance to split nodes. The importance of each variable will be ranked from the root to the top of the tree. For the AKH dataset, the variable importance determined by the DT and the RF algorithm has been depicted in Fig. 11. Because of the use of six *mtry* values to build the tree in the RF algorithm, the six most important parameters ranked by the RF method, including Coliform, DO, Turbidity, TSS, NH_4^+ −N, and BOD_5 are chosen. The same number of parameters in the use of the DT algorithm is also considered. Both algorithms agree that the Coliform is the most important parameter, followed by the DO. The Turbidity, BOD_5, and NH_4^+ −N are also the common features selected by the two methods, but with different levels of importance. In addition, while the RF selects TSS, the DT chooses BOD_5.

The performance of the ML models based on these parameters is

Table 5
The performance of ML models based on selected RF features and DT features (best results are in bold).

ML method	Selection method	Sim	R	RMSE	MAE	FSD	FA2	NSE
RF	RF	**0.939**	**0.90**	**11.07**	**6.52**	**0.09**	**0.94**	**0.77**
	DT	0.913	0.86	13.51	8.79	0.10	0.87	0.67
DT	RF	0.922	0.88	12.41	8.81	0.09	0.88	0.73
	DT	0.846	0.60	21.20	16.62	0.30	0.66	−0.21
SVM	RF	0.877	0.70	18.69	14.75	0.35	0.73	−0.03
	DT	0.849	0.62	20.98	16.23	0.25	0.66	−0.06
MLP	RF	0.879	0.71	18.35	14.47	0.36	0.75	−0.03
	DT	0.846	0.59	21.61	16.68	0.29	0.67	−0.22
LR	RF	0.854	0.60	20.84	17.66	0.48	0.68	−0.72
	DT	0.835	0.57	21.37	17.60	0.44	0.69	−0.67

Table 6
The performance of the RF model based on different selected RF features (best results are in bold).

Number of parameter	Name of parameter	ML method	Sim	R	RMSE	MAE	FA2	NSE
6	Coliform, DO	RF	**0.939**	**0.90**	**11.37**	**7.02**	**0.09**	**0.944**
	Turbidity	DT	0.92	0.88	12.41	8.81	0.09	0.88
	TSS	SVM	0.88	0.70	18.69	14.75	0.35	0.73
	NH_4^+ −N	MLP	0.85	0.60	20.84	17.66	0.48	0.68
	BOD_5	LR	0.88	0.71	18.35	14.47	0.36	0.75
5	Coliform	RF	**0.938**	**0.91**	**11.02**	**6.99**	**0.08**	**0.939**
	DO	DT	0.924	0.89	11.898	8.567	0.108	0.894
	Turbidity	SVM	0.876	0.688	18.910	14.844	0.344	0.728
	TSS	MLP	0.854	0.585	21.057	17.742	0.483	0.691
	NH_4^+ −N	LR	0.876	0.690	18.762	14.811	0.367	0.751
4	Coliform	RF	**0.940**	**0.910**	**10.78**	**6.80**	**0.079**	**0.945**
	DO	DT	0.924	0.89	11.90	8.57	0.11	0.89
	Turbidity	SVM	0.876	0.69	18.84	14.86	0.37	0.75
	TSS	MLP	0.854	0.59	21.02	17.67	0.49	0.69
		LR	0.876	0.68	19.04	14.90	0.35	0.74
3	Coliform	**RF**	**0.927**	**0.90**	**11.417**	**8.203**	**0.10**	**0.919**
	DO	DT	0.916	0.87	12.92	9.51	0.16	0.88
	Turbidity	SVM	0.871	0.66	19.55	15.59	0.39	0.71
		MLP	0.855	0.59	21.02	17.60	0.49	0.69
		LR	0.868	0.64	19.85	15.95	0.46	0.70

displayed in Table 5. It can be seen that the accuracy of the RF model in calculating the WQI is somewhat improved by using RF features. Indeed, based on six RF parameters, the RF model results in a similarity of 0.939, an MAE of 6.52, and an FSD of 0.09, slightly better than those 0.934, 6.78, and 0.16 of the same metrics obtained when using all ten parameters.

Table 5 also points out that the RF features are more effective than the DF ones. Values of the metrics corresponding to the use of the RF parameters are all better (i.e. either greater than for the Similarity, R, FA2, and NSE values, or less than for the RMSE, MAE, and FSD values) than the ones corresponding to the use of DF parameters. Even the use of DF parameters for the DT model (embedded itself) leads to worse performance than the use of RF parameters. For instance, the similarity of the DT model based on the DT features is 0.846, less than a value of 0.922 when using the RF features.

The obtained results shown in Table 5 enable us to draw a conclusion that the use of six RF features, including Coliform, DO, Turbidity, TSS, NH_4^+ −N, and BOD_5 for the ML models are effective. That is, the embedded selection method somewhat meets our expectations: it helps to reduce four water quality parameters without losing the accuracy of the ML models. However, according to the importance of each parameter shown in Fig. 11, we recognize that the importance of the next three parameters, namely BOD_5, NH_4^+ −N, and TSS, is not superior to the previous four parameters. This suggests us for conducting more experiments by eliminating them one by one and checking the performance of the ML algorithms in each case. For these experiments, we use only the RF parameters due to their superiority in the previous test. After getting rid of the BOD_5, and then, the NH_4^+ −N, we find that the ML models are still effective as the predicted values of WQI are close to the true values. However, when we discard one more parameter, which is TSS, the performance of all the ML methods is decreased. Therefore, it is reasonable to stop the experiments at this step. A summary of the experimental results is presented in Table 6. It shows that the best performance of the ML models is achieved when using four parameters, including Coliform, DO, Turbidity, and TSS. Using these four parameters, the RF model leads to a similarity of 0.940, an R score of 0.91, an MAE of 6.3, an FSD of 0.08, an FA2 of 0.94, and an NSE of 0.8. They are all better than the corresponding values of other cases.

5.3.4. *A visualization for the performance of the ML methods in predicting the WQI*

Besides making a quantitative evaluation of the performance of the ML methods in calculating the WQI, we also provide a visualization for the comparison between predicted WQI and the *true* WQI. In particular, Figs. 12 and 13 display the predicted WQI values given by the ML methods and the *true* WQI values calculated by the manual method. Fig. 12 shows the difference when using all 10 parameters for the ML models, while Fig. 13 is for the case of using only four RF parameters. In general, the results presented in the two figures are consistent with the ones discussed previously. The RF model shows the best capability in computing the WQI values: the predicted values are very close to the true values, especially the peaks. The WQI values predicted by the DT model are also close to the true values, however, when taking a closer look at the two figures, the model tends to give the same prediction for different values of input (the predicted values of the model form many straight lines in the two figures). The proximity between the WQI values given by the MLP model, the MLR model, and the SVM model and the true WQI values is less impressive. This indicates that the tree-based ML models are superior to the other model considered in this study. Finally, the use of four parameters selected by the RF algorithm leads to a better prediction of the WQI values. For instance, by taking a look at the highest peaks in the two figures, one can see that: the peak representing a true value of WQI is closer to the predicted WQI in Fig. 13 than the corresponding value in Fig. 12. This indicates the effectiveness of the selection method in improving the performance of the ML models.

5.4. *Discussion*

By using the embedded method for the feature selection combined with the RF algorithm, we have found that Coliform, DO, Turbidity, and TSS are the parameters that have the most influence on water quality at the An Kim Hai irrigation system. These four crucial parameters are also included in the WQI calculation worldwide. Indeed, Coliform is an important parameter representing the biological aspect of water quality. It is stipulated in water quality regulations of all countries over the world (Ott, 1978; Dunnette, 1979; Miller et al., 1986). DO is an essential water quality parameter that supports the survival of aquatic organisms and the self-purification of surface water. It is highly inversely correlated to BOD_5 and COD through the biological decomposition and NH_4^+

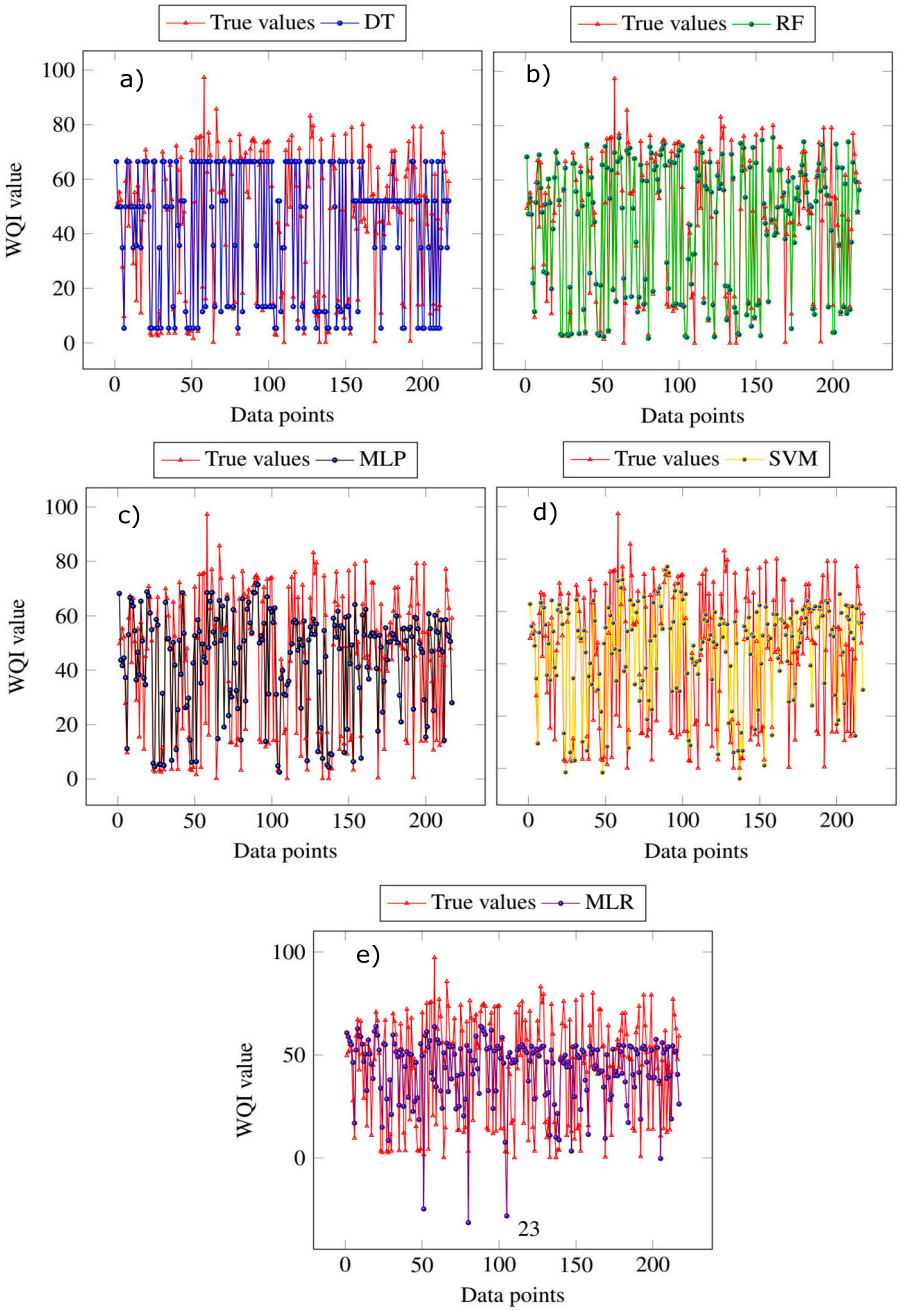


Fig. 12. A visualization for the difference between the predicted WQI values and the *true* WQI values using all features.

through the nitrification. Thus, DO can represent well other related parameters and it was widely used for the WQI calculation. For example, it has the highest weight of 0.17 in the NSF-WQI calculation (Ott, 1978; Dunnette, 1979; Miller et al., 1986). Meanwhile, Turbidity is an indicator reflecting the degree of land use in the watershed. Usually, it has been one of the few most crucial parameters for evaluating water quality (Pesce and Wunderlin, 2000; Debels et al., 2005; Wu et al., 2021). Finally, TSS refers to the dry weight of suspended particles in a sample of water. This parameter is considered a conventional pollutant in the U.S. Clean Water Act. Similar to the previous three parameters, TSS is stipulated in many studies for water quality assessment (Brown et al., 1970; Othman et al., 2020).

In comparison with the existing methods in the literature, our proposed method can directly and automatically find the key water quality parameters without trying many possible combinations. This is obviously more efficient than the "trial and error" technique that has been used in many existing studies (Bui et al., 2020; Ahmed et al., 2019; Khoi et al., 2022). The obtained results from this study show the potential application of ML-based methods for calculating the WQI in particular assessing water quality in general. However, there are some issues that should be kept in mind when applying these algorithms. Firstly, an ML-based model (combining feature selection and ML algorithms) should aim to reduce the number of water quality parameters in use. An ML model that requires the same input parameters as traditional methods does not make sense to be applied in practice. No matter how close the predicted WQI values given by this model are to the actual WQI values, it is only for reference. This is because the implementation of the traditional methods is straightforward and much simpler than the ML-based method. It can be done quickly even with Excel without wasting time and other resources for training and testing. Secondly, in this study,

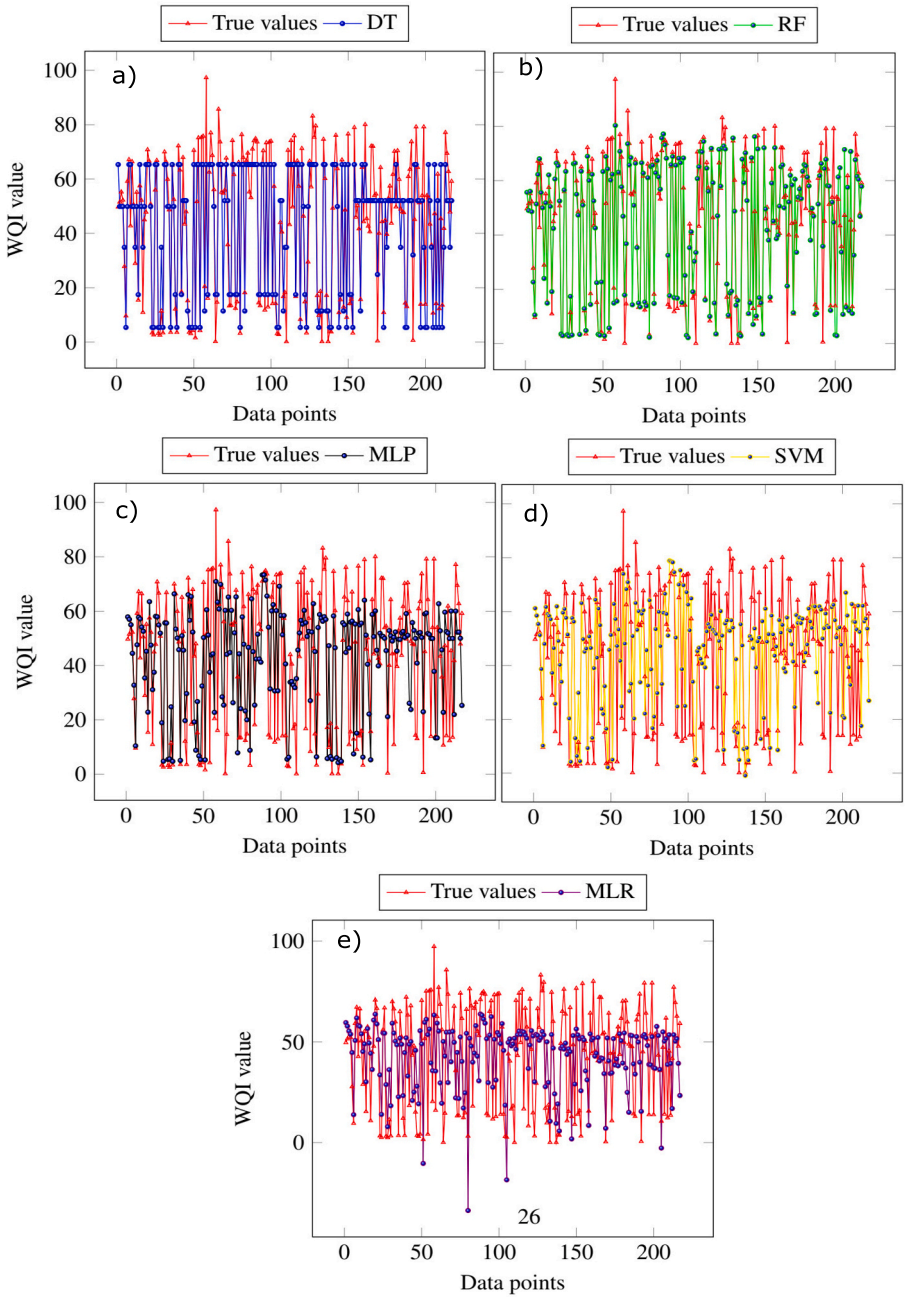


Fig. 13. A visualization for the difference between the predicted WQI values and the *true* WQI values based on four RF features.

based on the AKH dataset, we have successfully found the best model for evaluating the VN_WQI which combines the embedded feature selection method with RF algorithm. However, in other different datasets, these algorithms may no longer give the best performance. In these situations, the authors should conduct other experiments to choose the right feature selection and ML algorithms.

6. Conclusion

In this study, we have proposed a novel ML-based method for calculating the WQI of the AKH irrigation system in Vietnam. Several ML algorithms have been considered, including LR, MLP, SVM, DT, and RF. The feature selection methods to choose the key water quality parameters before feeding them to the ML models have also been applied. By applying these methods, the number of water quality parameters can be reduced significantly without losing the accuracy of the ML models. While the filter method and the wrapper method can reduce the number of parameters from 10 to 9 and 6, respectively, the embedded method with the RF features has figured out that using only four parameters, including Coliform, DO, Turbidity, and TSS, is enough for the RF model to compute the WQI accurately. This result has a great practical meaning. Indeed, in the implementation of ML models, using fewer parameters means less redundant data and misleading data. Accordingly, the model can be more accurate while training faster. From an economic point of view, it means saving much time and cost in measuring water quality parameters. In addition, it allows for avoiding the inaccurate measurement of some parameters that could lead to a final incorrect WQI value from the manual method. Several disadvantages of the traditional methods can be overcome by applying the ML methods. This finding, therefore, provides a promising alternative way to predict the WQI with a minimum number of water quality parameters in an irrigation system as well as other similar ones. For further research, we

Table 7
VN_WQI levels and suitability for use.

WQI values	Water quality	Suitable for use
91–100	Excellent	surface water can be used for the purpose of domestic water supply
76–90	Good	surface water can be used for a source of domestic water supply with appropriate treatment
51–75	Medium	surface water can be used for irrigation purposes
26–50	Bad	surface water can be used for other purposes that need lower water quality such as navigate
<25	Very bad	wastewater that can be discharged into the permitted water bodies for further

would like to explore other advanced ML models like ensemble algorithms or deep learning algorithms to improve the performance of calculating the WQI.

Funding

This research was funded by Thuyloi University Foundation for Science and Technology.

CRediT authorship contribution statement

Bui Quoc Lap: Conceptualization, Methodology, Writing-original-draft, Writing-review-editing. **Thi-Thu-Hong Phan:** Conceptualization, Methodology, Software, Writing-original-draft, Writing-review-editing, Investigation, Data-curation. **Huu Du Nguyen:** Investigation, Writing-original-draft, Writing-review-editing. **Le Xuan Quang:** Resources. **Phi Thi Hang:** Resources. **Nguyen Quang Phi:** Writing-original-draft. **Vinh Truong Hoang:** Conceptualization, Writing-original-draft. **Pham Gia Linh:** Resources. **Bui Thi Thanh Hang:** Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Acknowledgments

Water quality data were collected from the project “Monitoring water Quality in An Kim Hai Irrigation System, Supporting Water Intake for Agricultural Production” managed by the Ministry of Agriculture and Rural Development, Vietnam.

Appendix

References

Ahmed, A.N., Othman, F.B., Afan, H.A., Ibrahim, R.K., Fai, C.M., Hossain, M.S., Ehteram, M., Elshafie, A., 2019. Machine learning methods for better water quality prediction. J. Hydrol. 578, 124084.
Amral, N., Ozveren, C.S., King, D., 2007. Short term load forecasting using Multiple Linear Regression. In: 2007 42nd International Universities Power Engineering Conference, pp. 1192–1198.
Asadollah, S.B.H.S., Sharafati, A., Motta, D., Yaseen, Z.M., 2021. River water quality index prediction and uncertainty analysis: A comparative study of machine learning models. J. Environ. Chem. Eng. 9, 104599.
Bommert, A., Sun, X., Bischl, B., Rahnenführer, J., Lang, M., 2020. Benchmark for filter methods for feature selection in high-dimensional classification data. Comput. Stat. Data Anal. 143, 106839.
Brown, R.M., McClelland, N.I., Deininger, R.A., Tozer, R.G., 1970. A water quality index: Do we dare? Water Sewage Works 117, 339–343.
Bui, D.T., Khosravi, K., Tiefenbacher, J., Nguyen, H., Kazakis, N., 2020. Improving prediction of water quality indices using novel hybrid machine-learning algorithms. Sci. Total Environ. 721, 137612.
Chou, J.S., Ho, C.C., Hoang, H.S., 2018. Determining quality of water in reservoir using machine learning. Ecol. Inform. 44, 57–75.
Debels, P., Figueroa, R., Urrutia, R., Barra, R., Niell, X., 2005. Evaluation of water quality in the Chillán River (Central Chile) using physicochemical parameters and a modified water quality index. Environ. Monit. Assess. 110, 301–322.
Derdour, A., Jodar-Abellan, A., Pardo, M.Á., Ghoneim, S.S., Hussein, E.E., 2022. Designing efficient and sustainable predictions of water quality indexes at the regional scale using machine learning algorithms. Water 14, 2801.
Dunnette, D.A., 1979. A Geographically Variable Water Quality Index Used in Oregon. J. (Water Pollut. Control Fed.) 51, 53–61. Publisher: Water Environment Federation.
Fathi, E., Zamani-Ahmadmahmoodi, R., Zare-Bidaki, R., 2018. Water quality evaluation using water quality index and multivariate methods, beheshtabad river, iran. Appl. Water Sci. 8, 1–6.
Gupta, S., Gupta, S.K., 2021. A critical review on water quality index tool: Genesis, evolution and future directions. Ecol. Inform. 63, 101299.
Hameed, M., Sharqi, S.S., Yaseen, Z.M., Afan, H.A., Hussain, A., Elshafie, A., 2017. Application of artificial intelligence (AI) techniques in water quality index prediction: a case study in tropical region, Malaysia. Neural Comput. Appl. 28, 893–905.
Ho, J.Y., Afan, H.A., El-Shafie, A.H., Koting, S.B., Mohd, N.S., Jaafar, W.Z.B., Sai, H.L., Malek, M.A., Ahmed, A.N., Mohtar, W.H.M.W., Amin, E., Ahmed, E., 2019. Towards a time and cost effective approach to water quality index class prediction. J. Hydrol. 575, 148–165.
Jiang, J., Tang, S., Han, D., Fu, G., Solomatine, D., Zheng, Y., 2020. A comprehensive review on the design and optimization of surface water quality monitoring networks. Environ. Modell. Softw. 132, 104792.
Khoi, D.N., Quan, N.T., Linh, D.Q., Nhi, P.T.T., Thuy, N.T.D., 2022. Using machine learning models for predicting the water quality index in the la buong river, vietnam. Water 14, 1552.
Kim, S.E., Seo, I.W., Choi, S.Y., 2017. Assessment of water quality variation of a monitoring network using exploratory factor analysis and empirical orthogonal function. Environ. Modell. Softw. 94, 21–35.
Kocer, M.A.T., Sevgili, H., 2014. Parameters selection for water quality index in the assessment of the environmental impacts of land-based trout farms. Ecol. Ind. 36, 672–681.
Manzar, M.S., Benaafi, M., Costache, R., Alagha, O., Mu’azu, N.D., Zubair, M., Abdullahi, J., Abba, S., 2022. New generation neurocomputing learning coupled with a hybrid neuro-fuzzy model for quantifying water quality index variable: A case study from saudi arabia. Ecol. Inform., 101696
Miller, W., Joung, H., Mahannah, C., Garrett, J., 1986. Identification of water quality differences in Nevada through index application. Technical Report. Wiley Online Library.
Miller, W.W., Joung, H.M., Mahannah, C.N., Garrett, J.R., 1986. Identification of Water Quality Differences in Nevada Through Index Application. J. Environ. Qual. 15, 265–272.
Misaghi, F., Delgosha, F., Razzaghmanesh, M., Myers, B., 2017. Introducing a water quality index for assessing water for irrigation purposes: A case study of the ghezel ozan river. Sci. Total Environ. 589, 107–116.
Mitra, P., Murthy, C., Pal, S., 2002. Unsupervised feature selection using feature similarity. IEEE Trans. Pattern Anal. Mach. Intell. 24, 301–312 conference Name: IEEE Transactions on Pattern Analysis and Machine Intelligence.
Mohd Rosely, N.F.L., Salleh@Sallehuddin, R., Zain, A.M., 2019. Overview Feature Selection using Fish Swarm Algorithm. J. Phys.: Conf. Ser. 1192, 012068.
MONRE, 2019. The Decision on the promulgation of technical guidelines for calculation and publication of the Vietnam water quality index (VN_WQI).
Najah, A., El-Shafie, A., Karim, O.A., Jaafar, O., 2011. Integrated versus isolated scenario for prediction dissolved oxygen at progression of water quality monitoring stations. Hydrol. Earth Syst. Sci. 15, 2693–2708.
Nash, J.E., Sutcliffe, J.V., 1970. River flow forecasting through conceptual models part I — A discussion of principles. J. Hydrol. 10, 282–290.
Othman, F., Alaaeldin, M., Seyam, M., Ahmed, A.N., Teo, F.Y., Ming Fai, C., Afan, H.A., Sherif, M., Sefelnasr, A., El-Shafie, A., 2020. Efficient river water quality index prediction considering minimal number of inputs variables. Eng. Appl. Comput. Fluid Mech. 14, 751–763.
Ott, W., 1978. Environmental indices: Theory and practice. Ann Arbor Science, Ann Arbor, Mich.
Pesce, S.F., Wunderlin, D.A., 2000. Use of water quality indices to verify the impact of córdoba city (argentina) on suquıéa river. Water Res. 34, 2915–2926.
Pesce, S.F., Wunderlin, D.A., 2000. Use of water quality indices to verify the impact of Córdoba City (Argentina) on Suquıéa River. Water Res. 34, 2915–2926.
Ratner, B., 2009. The correlation coefficient: Its values range between+ 1/- 1, or do they? J. Target. Meas. Anal. Mark. 17, 139–142.
Rezaie-Balf, M., Nasrin, F., Razzaghmanesh, M., Mohammadzadeh, A., Murti, M.A., Ahmed, A.N., Fai, C.M., Nabipour, N., Alaghmand, S., El-Shafie, A., 2020. Physicochemical parameters data assimilation for efficient improvement of water quality index prediction: Comparative assessment of a noise suppression hybridization approach. J. Clean. Prod. 271, 122576.

Shamsuddin, I.I.S., Othman, Z., Abba, S.I., Abba, S.I., Pham, Q.B., Sani, G., Linh, N.T.T., Ahmed, A.N., Mohajane, M., Khaledian, M., Abdulkadir, R.A., Vu, Q.B., 2022. Implementation of data intelligence models coupled with ensemble machine learning for prediction of water quality index. Environ. Sci. Pollut. Res. 27, 41524–41539.

Shamsuddin, I.I.S., Othman, Z., Sani, N.S., 2022. Water quality index classification based on machine learning: A case from the langat river basin model. Water 14, 2939.

Shuhaimi-Othman, M., Lim, E.C., Mushrifah, I., 2007. Water Quality Changes in Chini Lake, Pahang, West Malaysia. Environ. Monit. Assess. 131, 279–292.

Song, L., Smola, A., Gretton, A., Borgwardt, K.M., Bedo, J., 2007. Supervised feature selection via dependence estimation. In: Proceedings of the 24th international conference on Machine learning - ICML '07. ACM Press, Corvalis, Oregon, pp. 823–830.

Sutadian, A.D., Muttil, N., Yilmaz, A.G., Perera, B.J.C., 2016. Development of river water quality indices-a review. Environ. Monit. Assess. 188, 58.

Sutadian, A.D., Muttil, N., Yilmaz, A.G., Perera, B.J.C., 2018. Development of a water quality index for rivers in West Java Province, Indonesia. Ecol. Ind. 85, 966–982.

Suwadi, N.A., Derbali, M., Sani, N.S., Lam, M.C., Arshad, H., Khan, I., Kim, K.I., 2022. An optimized approach for predicting water quality features based on machine learning. Wirel. Commun. Mob. Comput.

Uddin, M.G., Nash, S., Olbert, A.I., 2021. A review of water quality index models and their use for assessing surface water quality. Ecol. Ind. 122, 107218.

Uddin, M.G., Nash, S., Rahman, A., Olbert, A.I., 2022. A comprehensive method for improvement of water quality index (wqi) models for coastal water quality assessment. Water Res. 219, 118532.

Uddin, M.G., Nash, S., Rahman, A., Olbert, A.I., 2022. A novel approach for estimating and predicting uncertainty in water quality index model using machine learning approaches. Water Res. 119422.

Uddin, M.G., Nash, S., Rahman, A., Olbert, A.I., 2022. Performance analysis of the water quality index model for predicting water state using machine learning techniques. Process Saf. Environ. Prot.

Vapnik, V.N., 1995. The Nature of Statistical Learning Theory. Springer-Verlag, Berlin, Heidelberg.

Wu, Z., Wang, X., Chen, Y., Cai, Y., Deng, J., 2018. Assessing river water quality using water quality index in lake taihu basin, china. Sci. Total Environ. 612, 914–922.

Wu, Z., Lai, X., Li, K., 2021. Water quality assessment of rivers in lake chaohu basin (china) using water quality index. Ecol. Ind. 121, 107021.

Yan, K., Zhang, D., 2015. Feature selection and analysis on correlated gas sensor data with recursive feature elimination. Sens. Actuators: B. Chem. Compl. 353–363.